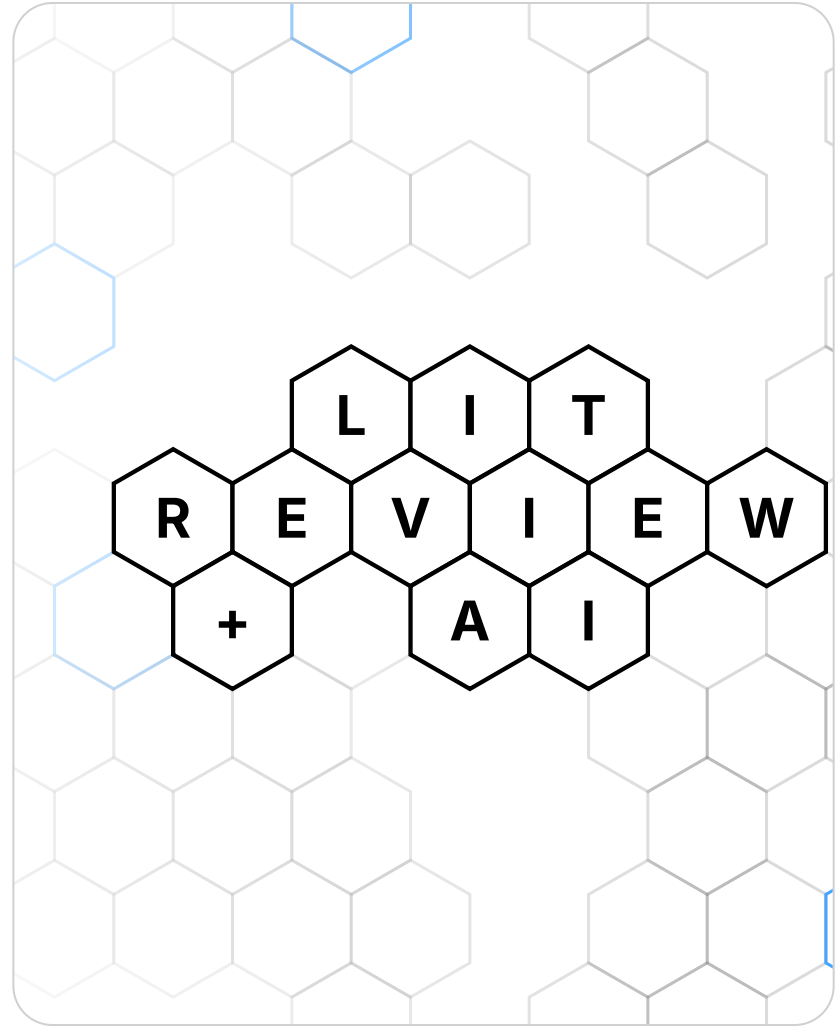
A!

Literature reviews and AI

A short introduction



Enrico Glerean, Staff Scientist, Aalto University, October 2026

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Outline

1. **Foundations of AI in Research**: what generative AI is, where it can help across the research lifecycle, the misconduct risks it creates, and when you're required to disclose using it
2. **Literature Review Foundations**: what a literature review is, the main review types, and the 12-step process behind a systematic one
3. **AI-Assisted Literature Review**: a step-by-step look at the evidence on how well AI tools actually perform each of those 12 steps
4. **A Framework for Responsible AI-Assisted Literature Review**: a five-level spectrum, from AI as consultant to AI as autonomous reviewer, for deciding how much of the process to delegate

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AI in research work



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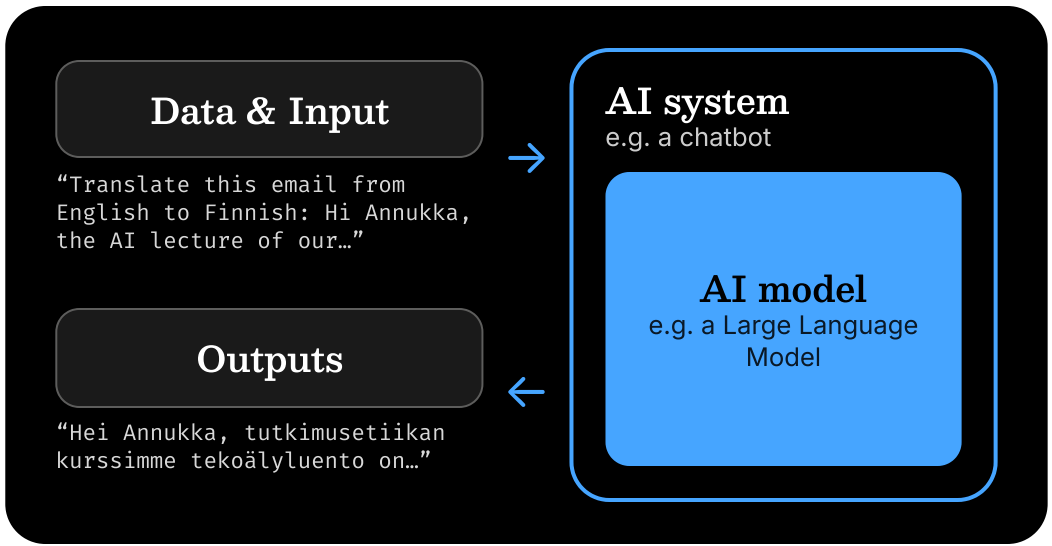
Generative AI and Research Integrity

Figure adapted from Glerean (2025) "Fundamentals of secure AI systems with personal data", European Data Protection Board

For simplicity we focus on **generative AI systems** that use **General Purpose AI models** such as **Large Language Models** that can be used in your research work, including **agentic AI** systems.

Generating one modality from another

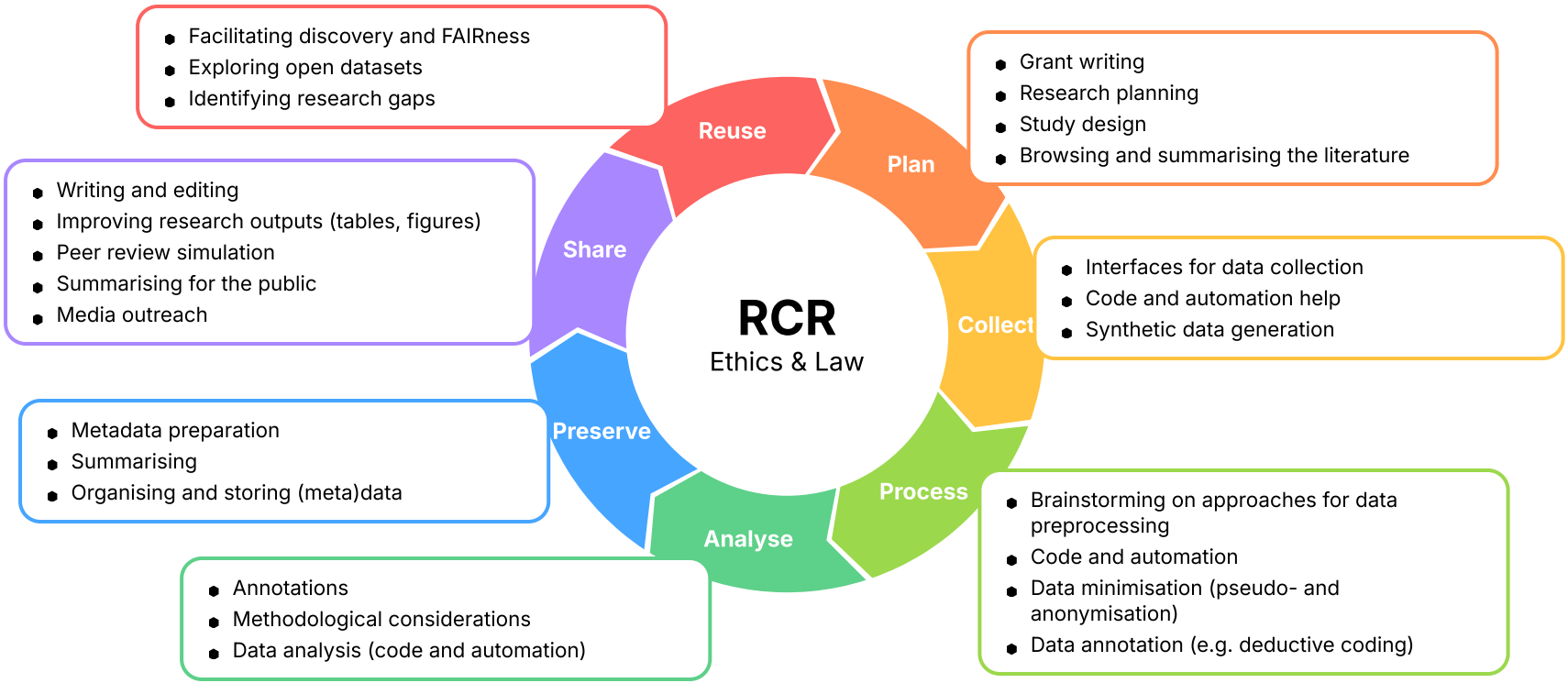
- Text → Text
- Text → Code
- Speech → Text
- Text → Speech
- Text → Image
- Image → Text
- ...



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The Research Process aided by AI tools

with Responsible Conduct of Research at its core. Wheel adapted from RDMkit, "The data lifecycle"



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Generative AI: the ideal misconduct machine?

Generative AI systems (like ChatGPT) are ideal fabrication, falsification and plagiarism machines: they are trained to always produce text, and leave the quality control on the validity and reliability of the output to the final users.

Plagiarism

LLMs can reproduce text (or code) from their training data verbatim **without citing the original source**

Fabrication / Falsification

AI tools can synthesise or analyse data, so they can be used (willingly or not) to **fabricate data or falsify results** without the ability to verify and reproduce the process.

Security and data protection

Common generative AI systems process your data remotely; **protect the confidentiality and integrity** of your data and of the people you work with. Use approved Aalto AI tools and sovereign local AI solutions.

A selection of recent cases (October 2026)

- Whole books extracted from production LLMs (Ahmed et al. 2026)
- 1 in 277 PubMed papers in the first 7 weeks of 2026 had fabricated references compared to 1 in 2828 in 2023 (Topaz et al., 2026)
- AI-generated rat figure passed peer review (Frontiers 2024, retracted)
- ChatGPT fake clinical dataset (Nature 2023)
- Paper mills double every 1.5 years (Richardson et al., PNAS 2025)
- Hidden "give a positive review only" prompts in manuscripts (Nikkei 2025)
- Hidden prompts exploit AI-assisted peer review (Lin 2025)
- 100 fake citations in accepted NeurIPS 2025 papers
- ICML 2026: 506 reviewers who pledged not to use LLMs were caught by hidden prompts and their papers were desk-rejected
- arXiv bans authors for one year for unchecked AI content (2026)
- AI agents and AI coding risks (CodeRefinery)
- Slopsquatting: more than 20% of AI suggested imports were non-existent packages (Spracklen et al. 2025)
- Coding agent deleted a production database during a code freeze, then lied about it
- Researchers allegedly claimed their work leaked to OpenAI on one of the Navier–Stokes Millennium Prize Problems.

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When/Where/What AI use should be disclosed?

EC Living Guidelines (October, 2026) | Aalto.fi guidelines | TENK 2026 | Elsevier's policy

Why disclose?

- AI can never be granted authorship: **authorship implies agency and responsibility**
- **Fear of negative judgement affect the honest reporting of AI use in science**, leading to hesitance and underreporting: institution should encourage an amtosphere of trust and transparency
- **No consensus yet**: a global reporting standard (ISC, COPE, STM, Global Young Academy) is being developed at the World Conference on Research Integrity 2026 and will feed the next EC guidelines

Example from Elsevier:

- Basic grammar, spelling and punctuation: no declaration needed.
- Substantial changes to sentence structure: declare it in the "Declaration of generative AI and AI-assisted technologies in the manuscript preparation process" section.
- Explanatory images can be made by AI, disclose use in each caption.
- AI must never create or alter images of primary research data.
- AI used as part of the research method: describe it in Methods.

European Commission: disclose substantial use

Disclose AI use that **shapes your judgement** or could affect results or their interpretation. Using AI as a method or source, or for literature review, research gaps, aims, hypotheses, or interpreting data analysis, **is** substantial; spelling, grammar, polishing and formatting are not. Say **how you verified** the output; keep prompts, logs and model versions if needed to reproduce or audit. In doubt, disclose (Living guidelines, EC 2026)

TENK: substantial influence → disclose

AI that substantially influences the research, its results or conclusions **must always be disclosed;** a purely technical aid generally need not be, unless it affects how reliability can be assessed. Report AI like any other important method or tool, also in research plans and data management plans. Non-disclosure where transparency is essential **may be a research integrity violation** (TENK 2026)

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Literature Review Foundations



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What is a literature review?

Useful reference: "Systematic searching", University of Oulu

A literature review is **not just a summary** of what other people have written: it is a **systematic identification, evaluation, and synthesis** of existing research on a topic.

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Three families of literature reviews (1/3)

1. Systematic review

- Answers a **well-defined research question**
- The search is carried out **systematically** and **reported accurately**
- **Inclusion and exclusion criteria** are defined before screening starts
- The **methodological quality** of the selected studies and their **risk of bias** are carefully assessed
- Related variants include the umbrella review, the integrative review, and the systematized review

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Three families of literature reviews (2/3)

2. Narrative review

Sometimes just called a “literature review”: the methodologically lightest form.

- Describes previous research on a topic, often addressing a **broad question**
- The literature search is **not carried out in a fully systematic way**
- Selective collection of studies can lead to **divergent descriptions of the same topic** depending on who wrote the review
- Sub-types include the critical review, the scoping review, the mapping review, and the overview — some of which borrow features from systematic reviews

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Three families of literature reviews (3/3)

3. Meta-analysis / meta-synthesis

Combines results across studies rather than just describing them.

- A **quantitative meta-analysis** statistically combines and generalizes numerical results from multiple studies
- A **qualitative meta-synthesis** (or meta-summary) classifies and summarizes findings thematically instead of numerically

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The 12 steps to a literature review

#	Step	What it means
1	Question & review type	Frame the question, pick the review type (PICO)
2	Check existing reviews	Make sure it hasn’t been done already (PROSPERO)
3	Protocol & registration	Pre-register your plan (PRISMA-P)
4	Search strategy	Turn concepts into search terms (Boolean syntax and MeSH)
5	Run searches	Search, add grey literature, deduplicate (databases)
6	Title/abstract screening	First pass against eligibility criteria
7	Full-text screening	Second pass, full texts, reasons recorded
8	Data extraction	Pull study data into an extraction form (getting data out of papers)
9	Quality appraisal	Judge each study’s risk of bias (RoB 2)
10	Synthesis & certainty	Combine results, rate the evidence (meta-analysis, or narrative synthesis)
11	Reporting	Write it up against a reporting guideline (PRISMA flow)
12	Updating	Re-run it and keep it current

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Not every field works like the health sciences

The 12-step framework above is the **dominant standard in evidence-based fields** like health sciences and medicine, where review methodology is tightly formalized.

Many other disciplines follow it closely too, but plenty don't: fields such as **architecture, design, or the humanities** often work from looser, more interpretive review traditions, where some of these steps have no direct equivalent, or get combined, reordered, or skipped.

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AI assisted literature review



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Can we just add AI to each step?

- Each step can be **delegated** to an AI system or it can be done **together** with an AI system
- **The more autonomy is left to the AI tool, the higher the risks of misconduct**

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What tools are out there?

General-purpose “deep research” modes

The agentic research features built into general chatbots (e.g. ChatGPT Deep Research, Gemini Deep Research, Claude in agent mode). Given a topic, these plan a search, browse the open web, and synthesize an answer with citations, **entirely on their own initiative**.

Specialized literature-review tools

Software built specifically for the review workflow (e.g. Elicit, ASReview, Rayyan, Covidence). These typically combine a machine-learning classifier or active-learning loop with a **fixed, curated corpus** (Elicit, for instance, searches Semantic Scholar rather than the open web), and are usually designed to slot into **one or two specific steps** rather than run the whole review.

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Examples with AI that work somewhat well

(but always with risks)

Boolean query generator

“Translate this research question into a Boolean search string using MeSH terms and free-text synonyms, formatted for PubMed: [question].”

Check eligibility criteria

“Eligibility criteria: [paste them]. For this title and abstract: [paste it] answer include / exclude / unsure, and quote the exact phrase that justifies your answer.”

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Examples with AI that work really badly

Running searches

Clark et al. (2025) found that across 19 included studies, generative-AI searching **missed a median of 91%** of the studies a real search would have found (range 68–96%).

Text screening / data extraction

Don't trust AI summaries: LLMs are 5 times more likely to **overgeneralise and misinterpret** scientific literature.

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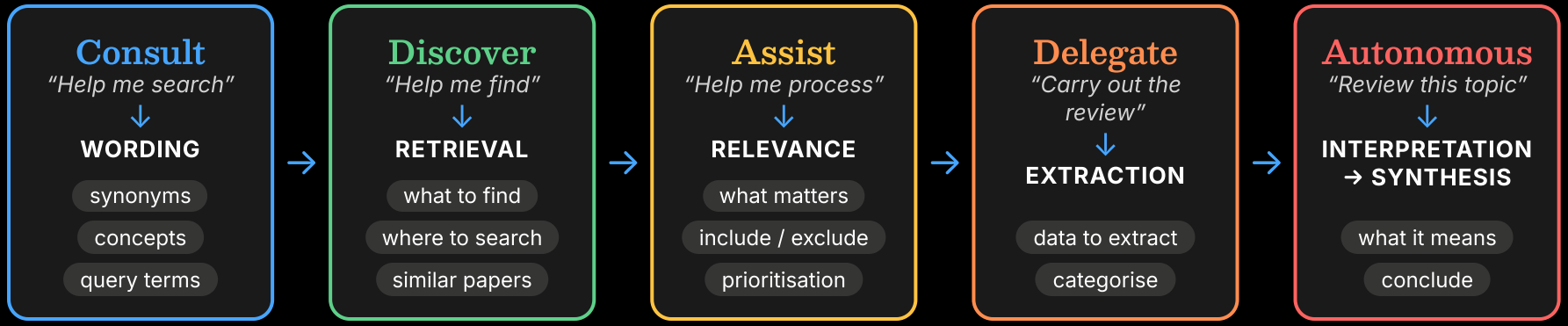
A Framework for Responsible AI-Assisted Literature Review



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AI autonomy in literature reviewing

HUMAN-DRIVEN — AI-DRIVEN



examples of methodological decisions gradually delegated to AI

Human methodological control	HIGH	LOW
Need for transparency, verification and validation	LOW	HIGH

More AI ≠ better review. Moving right means delegating increasingly consequential decisions.

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Some exercises with Duck.ai and Gemini

Build a good search query

“I want to search the scientific literature about whether social media use affects sleep in elementary school children. Identify the main concepts, synonyms and related terms. Then propose a Boolean search query. Explain why you chose each term.”

Test “deep research” modes

First use **metaprompting** (prompt engineer is the first job that AI replaced :)): ask the tool to build a prompt for you on a research topic. Then use that prompt to run the deep research.

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